



**Sir Syed University of Engineering & Technology**  
 Faculty of Computing and Applied Sciences (For CS, SE, IT, BI, MA,  
 Psychology)  
 Department of Computer Science & Information Technology

**Mid Term Examinations (Spring 2023)**

|                        |                                                   |          |          |                                     |
|------------------------|---------------------------------------------------|----------|----------|-------------------------------------|
| Course Code with Title | CS-467: Software Project Management               |          | Program  | Bachelor of Computer Science (BSCS) |
| Instructor             | Engr. M.Nadeem, Ms. Anam Siddiqui & Mr. M. Faheem |          | Semester | 7 <sup>th</sup>                     |
| Start date & Time      | May 19, 2022 at 11:30 AM                          | End Time | 01:00 PM |                                     |
| Maximum Marks          | 30                                                |          |          |                                     |

**IMPORTANT INSTRUCTIONS:**

Read the following Instructions carefully:  
 Attempt All Questions.

**Q.1.**

**(06)**

Suppose ABC Infrastructure, a manufacturing company, uses the expected monetary value (EMV) method to analyze a project's risk score and develop a contingency reserve. Calculate contingency reserve by using following table.

| Risk-ID | Risk                 | Probability | Impact    |
|---------|----------------------|-------------|-----------|
| 1       | Risk A (Threat)      | 12%         | \$24,000  |
| 2       | Risk B (Threat)      | 58%         | \$15,000  |
| 3       | Risk C (Threat)      | 9%          | \$54,000  |
| 4       | Risk D (Threat)      | 18%         | \$21,000  |
| 5       | Risk E (Opportunity) | 23%         | \$-22,000 |
| 6       | Risk F (Threat)      | 65%         | \$72,000  |

Q1  $EMV = \text{Probability} \times \text{Impact}$

| RiskId | Risk                 | Prob | Impact    |
|--------|----------------------|------|-----------|
| 1      | Risk A (Threat)      | 12%  | \$24,000  |
| 2      | Risk B (Threat)      | 58%  | \$15,000  |
| 3      | Risk C (Threat)      | 9%   | \$54,000  |
| 4      | Risk D (Threat)      | 18%  | \$21,000  |
| 5      | Risk E (Opportunity) | 23%  | -\$22,000 |
| 6      | Risk F (Threat)      | 65%  | \$72,000  |

EMV

$$0.12 \times 24000 = 2880$$

$$0.58 \times 15000 = 8700$$

$$0.09 \times 54000 = 4860$$

$$0.18 \times 21000 = 3780$$

$$0.23 \times -22000 = -5060$$

$$0.65 \times 72000 = 46800$$

Contingency Reserve = Sum of all EMVs

$$= 2880 + 8700 + 4860 + 3780$$

$$- 5060 + 46800$$

$$= 61690. \text{ Ans.}$$

Q.2.

(06)

ABC software house will initially investment \$20000 for its software project with discount rate is 14%. Project represents a series of cash flows as follows

| Year | Cash Inflows |
|------|--------------|
| 1    | \$ 2,300     |
| 2    | \$ 3,300     |
| 3    | \$ 3,200     |
| 4    | \$ 6,400     |
| 5    | \$ 3,500     |

- Compute net present value (NPV) of the project.
- Is project acceptable to company? Why

Q2 Initial investment = \$20,000  
Discount rate = 14% = 0.14

| Year | Cash Inflows |
|------|--------------|
| 1    | \$2,300      |
| 2    | \$3,300      |
| 3    | \$3,200      |
| 4    | \$6,400      |
| 5    | \$3,500      |

(a) Year 1:

$$\begin{aligned} PV &= \frac{CF}{(1+r)^n} \\ &= \frac{2300}{(1+0.14)^1} \end{aligned}$$

$$= \frac{2300}{1.14}$$

$$PV = 2017.54$$

Year 2:

$$PV = \frac{CF}{(1+r)^n}$$

$$= \frac{3300}{(1+0.14)^2}$$

$$= \frac{3300}{1.2996}$$

$$PV = 2539.24$$

Year 3:

$$\begin{aligned} PV &= \frac{CF}{(1+r)^n} \\ &= \frac{3200}{(1+0.14)^3} \\ &= \frac{3200}{1.481544} \end{aligned}$$

$$PV = 2159.90$$

Year 4:

$$\begin{aligned} PV &= \frac{CF}{(1+r)^n} \\ &= \frac{6400}{(1+0.14)^4} \\ &= \frac{6400}{1.6889} \\ &= 3789.31 \end{aligned}$$

Year 5:

$$PV = \frac{CF}{(1+r)^n}$$

$$= \frac{3500}{(1+0.14)^5}$$

$$= \frac{3500}{1.9254}$$

$$= 1817.79$$

NPV = Sum of all PVs - Initial investment

$$= 12323.78 - 20000$$

$$= -7676.22 \quad \text{Ans}$$

(b) The project is not acceptable to the company because the net present value is negative.

Q.3.

(06)

You have taken over a software development project 10 months after it started, and need to find out project stakeholders and roles. What kind of document do you need and why? Discuss it?

### Key Document Needed:

The **Project Charter** or **Stakeholder Register** is required.

### Why These Documents?

#### 1. Project Charter:

- This document contains high-level information about the project, including:
  - Key stakeholders and their roles.
  - Objectives and scope of the project.
  - Authorization of the project.
- It is created at the beginning of the project and acts as a reference for project governance.

#### 2. Stakeholder Register:

- Specifically lists:
  - All stakeholders involved in the project.
  - Their roles and responsibilities.
  - Levels of influence and interest in the project.
- Helps in understanding who needs to be involved or consulted.



### Discussion:

Both documents are critical to re-establish project clarity:

- **Project Charter** gives an overview and foundational knowledge, helping you understand the project's scope and purpose.
- **Stakeholder Register** provides detailed, actionable information about who is involved and their specific roles, helping to streamline communications and decision-making.

Q.4.

(06)

PMI has developed a project management framework which provides a basic structure and knowledge areas to describe project management body of knowledge and practices in terms of their component processes. These processes have been organized into 10 knowledge areas. List down the each knowledge area with brief description?

### **1. Project Integration Management**

- **Description:** Focuses on ensuring all project elements are coordinated. It involves creating the project charter, developing the project management plan, managing and directing the work, and monitoring project progress.

### **2. Project Scope Management**

- **Description:** Ensures the project includes all the work required and only the work required to complete the project successfully. It involves scope planning, definition, validation, and control.

### **3. Project Schedule Management**

- **Description:** Deals with the development and management of the project schedule. This includes defining activities, sequencing them, estimating durations, and monitoring the project timeline.

### **4. Project Cost Management**

- **Description:** Focuses on planning, estimating, budgeting, and controlling project costs to ensure the project is completed within the approved budget.

### **5. Project Quality Management**

- **Description:** Ensures that the project will satisfy the requirements for which it was undertaken. It involves quality planning, quality assurance, and quality control processes.

### **6. Project Resource Management**

- **Description:** Focuses on identifying, acquiring, and managing the resources (people, equipment, materials) needed for successful project completion. It also includes managing team dynamics.

### **7. Project Communications Management**

- **Description:** Involves planning, distributing, and monitoring project information. Ensures effective communication among stakeholders throughout the project lifecycle.

## 8. Project Risk Management

- **Description:** Focuses on identifying, analyzing, and responding to project risks. It ensures that risks are minimized and opportunities are maximized during the project.

## 9. Project Procurement Management

- **Description:** Involves acquiring goods and services from external suppliers to complete the project. Includes planning procurements, selecting vendors, and managing contracts.

## 10. Project Stakeholder Management

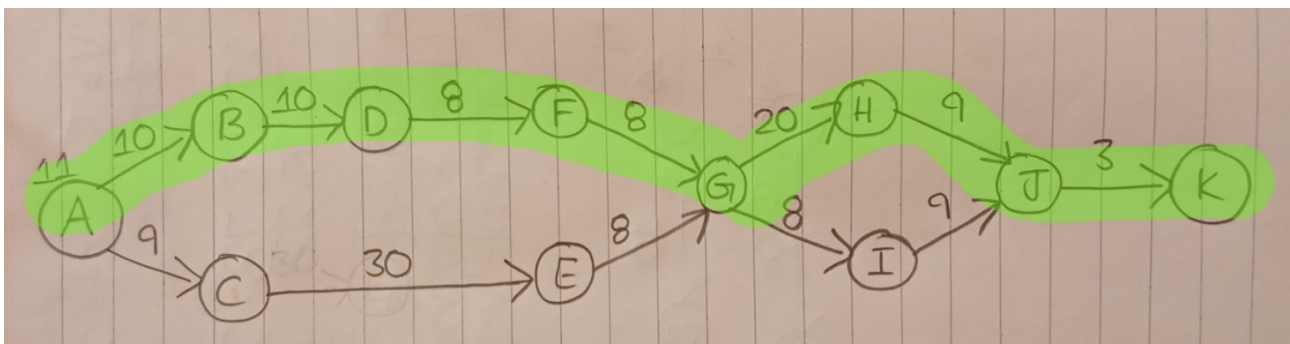
- **Description:** Focuses on identifying all stakeholders, analyzing their needs and expectations, and ensuring they are engaged throughout the project to meet their objectives.

Q.5.

(06)

Draw the project network diagram (PDM) and find the critical path by using following project plan.

| Activity | Predecessor | Duration in (Days) |
|----------|-------------|--------------------|
| A        | None        | 11                 |
| B        | A           | 10                 |
| C        | A           | 9                  |
| D        | B           | 10                 |
| E        | C           | 30                 |
| F        | D           | 8                  |
| G        | E & F       | 8                  |
| H        | G           | 20                 |
| I        | G           | 8                  |
| J        | H & I       | 9                  |
| K        | J           | 3                  |



Paths:-

$$A \rightarrow B \rightarrow D \rightarrow F \rightarrow G \rightarrow H \rightarrow J \rightarrow K = 11 + 10 + 10 + 8 + 8 + 20 + 9 + 3 = 79$$

$$A \rightarrow C \rightarrow E \rightarrow G \rightarrow I \rightarrow J \rightarrow K = 11 + 9 + 30 + 8 + 8 + 9 + 3 = 78$$